

3.8 Problem set 3

Problem 3.1: Unitary operators and states normalisation

Let $\hat{U}$ be a unitary operator, and let $|\psi\rangle$ be the state of a quantum system. Show that this unitary preserves normalisation, i.e.,

$$\| |\psi\rangle \| = \| \hat{U} |\psi\rangle \| = 1 .$$

Problem 3.2: Unitarity of Pauli operators

Show that Pauli operators, $\hat{X}$, $\hat{Y}$, and $\hat{Z}$ are unitary.

Problem 3.3: Eigenvalues and eigenvectors of Pauli operators

Find the eigenvalues and normalised eigenvectors of Pauli operators, $\hat{X}$, $\hat{Y}$, and $\hat{Z}$.

Problem 3.4: Square of Pauli matrices

1. Show that, for $\hat{A} \in \{\hat{X}, \hat{Y}, \hat{Z}\}$,

$$\hat{A}^2 = \mathbb{1} .$$

2. Use the previous result and show that

$$e^{i\theta\hat{A}} = \cos \theta \mathbb{1} + i \sin \theta \hat{A} .$$

3. We define the vector of Pauli matrices $\vec{\sigma}$ and the unit vector $\vec{n}$ by

$$\vec{\sigma} = \begin{pmatrix} \hat{X} \\ \hat{Y} \\ \hat{Z} \end{pmatrix} ,$$

$$\vec{n} = \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix} .$$

- (a) Prove that

$$\left(\vec{n} \cdot \vec{\sigma} \right)^2 = \mathbb{1} .$$

- (b) Deduce the expression of the operator $\hat{R}_\theta$, defined by

$$\hat{R}_\theta = e^{\frac{-i\theta}{2} \vec{n} \cdot \vec{\sigma}} .$$

Problem 3.5: Eigenstates of Hadamard gate

Show that the eigenstates of the Hadamard gate are given in the computational basis as

$$|v_{\pm}\rangle = \cos \frac{\pi}{8} |0\rangle \pm \sin \frac{\pi}{8} |1\rangle .$$

Problem 3.6: Circuit identities

Prove the following identities:

1.

$$\hat{H}\hat{X}\hat{H} = \hat{Z} .$$

2.

$$\hat{H}\hat{Y}\hat{H} = -\hat{Y} .$$

3.

$$\hat{H}\hat{Z}\hat{H} = \hat{X} .$$

Problem 3.7: Controlled gates

Find the matrix representations of the following controlled gates:

1. controlled- $\hat{Y}$.

2. controlled- $\hat{Z}$.

3. controlled- $\hat{H}$.

Problem 3.8: Quantum teleportation

We consider the quantum circuit shown in **Figure 3.6**. Show that $|\psi'\rangle = |\psi\rangle$.

In the aforementioned circuit, the state φ is the EPR state given by $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.